Case Study: National Youth Tobacco Study

The National Youth Tobacco Survey (NYTS) is a complex survey carried out by the CDC to better understand the rate of e-cigarette usage among teens. It is a school-based, pencil-and-paper questionnaire, self-administered to a nationally representative sample of students in grades 6-12 in the U.S. In 2019, 5,675 students completed the NYTS; the overall survey response rate was 71.6%. Among other questions, students were asked if they had used an e-cigarette in the past 30 days.

1. What is the population? Describe it in words.
2. How large is the population?
3. What is the sample? Describe it in words.
4. How large is the sample that the researchers obtained (what is n)? How large was it intended to be?
5. What is the unit of observation in the survey data?
6. Identify a population parameter that researchers would be interested in estimating.
7. Identify the corresponding statistic that will serve as the estimate of the population parameter you identified.
8. Identify possible sources of statistical bias and whether they are selection, measurement, and non-response bias. How do you expect this will affect the estimate of the parameter?
9. Identify possible sources of variation and whether they are sampling or measurement variability. If the CDC conducted multiple samples, how do you expect this will affect the closeness of the statistics in each sample to one another?

10. Sketch a *possible empirical distribution* of the NYTS data (a plot of the distribution of student responses).

- Be sure to label your axes.

11. Now, sketch a *possible sampling distribution* of the statistic you identified in **Question 7** given what you know about the data collection.

- Be sure to label your axes.
- Illustrate any anticipation of bias you have by adding a vertical line on the sampling distribution that hits the x-axis where you expect the population parameter to be.